DEPARTMENT OF MECHANICAL ENGINEERING

Small-scale two-stroke, compression ignition engine – Validation Report

Mechanical Design A (04 23808)

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1. PRODUCT DESIGN SPECIFICATION (PDS)

Key Consideration	Requirements	Issues / Challenges
Product Description	Small-scale 0.76 cc two-stroke CI UAV engine; 12,000 rpm; 15-min mission capability.	
Performance	12,000 rpm; 0.76 cc displacement; Fuel use: 11.2 mL/15 min; CR (start 8:1 & run 18:1);	Thermal efficiency at high rpm .
Materials & Process	AISI 4340 steel crankcase, crankshaft, piston; Steel liner; SKF 608-SZ bearings (8×22×7 mm).	Machining hardness of 4340 steel.
Dimensions	Bore: 10.49 mm; Stroke: 8.74 mm; h(start): 1.249 mm; h(run): 0.514 mm; Port widths: Intake 8.80 mm, Exhaust 1.59 mm, Transfer 2.50 mm; Height ~30 mm.	Port micro-machining accuracy; tolerance stacking.
Quantity	Initial manufacturing run: 50,000 units; Target: 1,000,000 in 5 years.	Scaling supply chain for steel & bearings.
Lifespan / Reliability	≥15 min flight per mission; Stress-checked components (SF ≥ 2).	Component wear at 12,000 rpm.
Maintenance	Fuel check per flight; Clean fins/ports every 10–20 h; Replace piston rings every 50 h.	
Aesthetics & Ergonomics	Compact form, minimal protrusions, clean mechanical layout. Quick-release mount; alignment marks.	Must not obstruct airflow or cooling.
Safety & Standards	Shaft guarding; exhaust insulation; safe fuel routing; BS 8888; Noise ≤ 75 dB @1 m.	Hot exhaust and high-rpm shaft.
Testing	Noise, vibration, thermal stress, bearing reliability.	
Manufacturing	CNC machining; micro-port cutting; 4340 heat-treat and finishing.	Tool wear and dimensional drift.
Packaging & Shipping	Shock-isolating packaging; corrosion-protected interior; compliant with fuel-system shipping laws.	Vibration loads during shipping.
Target Cost / Price	Final cost TBD after supplier evaluation and bulk quotes.	Price fluctuation of specialty steel.
Patents	Verify no conflict with two-stroke UAV engine patents.	Time of application.
Customers	UAV manufacturers, competition teams, research labs.	

2. ENGINE DESIGN SUMMARY.

The small-scale, two-stroke Compression Ignition (CI) engine is specifically designed for UAV exploration missions. It employs a side-port, short-stroke (oversquare) configuration to achieve high engine speeds while maintaining compact dimensions and low weight. The air-cooled engine delivers approximately 45 W (0.06 bhp) at 12,000 rpm and is optimized to drive a 200 × 100 pitch propeller, providing sufficient thrust for a minimum 15-minute flight duration.

- **Engine Parameters**

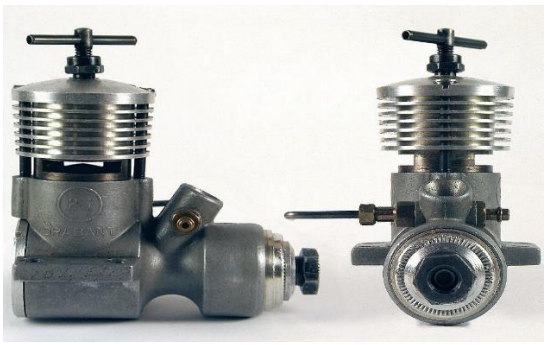
(Table 2: Engine Parameters)

Engine parameter	Value	Unit
Weight	50-100	g
Stroke / bore ratio	1.2	–
Stroke length	8.74	mm
Bore	10.49	mm
Displacement	755.36	mm^3
Total Fuel tank capacity	11.2	mL
Power output	45	W
Speed	12,000	rpm

Aesthetic and Ergonomic Features.

The integration of aesthetic and ergonomic considerations is crucial for the overall success and practicality of the small-scale, two-stroke Compression Ignition (CI) engine designed for Unmanned Aerial Vehicle (UAV) applications.
Ergonomic design plays a vital functional role by ensuring the engine is engineered for user comfort, safe operation, and ease of maintenance. Specific ergonomic features guarantee simple handling and installation, which are essential requirements for field deployment and quick maintenance turnaround in demanding environments where reliability is paramount.
Aesthetics serve to enhance the engine's professional appearance and overtly reflect high build quality. Collectively, these design features contribute to a user-friendly and durable propulsion system, confirming its suitability for rigorous UAV mission profiles.

Aspect	Design Intent and Description
User Handling	The engine features compact dimensions and contoured edges, ensuring safe and comfortable manual handling during installation and maintenance.
Component Accessibility	Mounting bolts, fuel inlets, and electrical connectors are strategically positioned to allow quick access, minimizing effort and hand strain for the operator.
Weight Distribution	The engine’s center of gravity is optimized for balanced lifting and smooth integration within the UAV fuselage.
Thermal Safety	Key surfaces are insulated and clearly marked with “safe touch” zones to protect operators from burns during post-flight maintenance.
Operator Feedback	Visual cues and alignment markings facilitate correct installation, reducing assembly time and operator fatigue.





DELMO 2.65 cc diesel

Engine/ Airframe mounting details.

Engine mounts play a critical role in securing the engine to the UAV airframe while mitigating vibrations, shocks, and noise. By absorbing these forces, the mounts reduce structural stress on the airframe, improving flight stability and overall performance.

Lightweight yet strong materials, such as aluminum and composites, are commonly used for the structural components of engine mounts. To further dampen engine vibrations, elastomeric materials like polyurethane are employed. These isolators efficiently absorb vibrational energy, protecting the airframe from fatigue and enhancing the durability of the propulsion system. Recent advances in elastomer technology have improved their damping capabilities, thermal resistance, and long-term reliability, making them ideal for aerospace applications.
In two-stroke engine configurations, elastomer bushings act as passive vibration isolators, preventing engine oscillations from being transmitted to the airframe. This ensures that both stiffness and damping are optimized to minimize structural fatigue over repeated flight cycles.

To maintain operational safety and reliability, engine mounts incorporate features such as locknuts, safety wires, and load-rated fasteners to prevent loosening under vibration and cyclic loads. Relief mechanisms are also integrated to handle dynamic forces during flight, ensuring that the engine remains securely mounted and the UAV operates smoothly.

Maintenance Schedule.

Regular maintenance is essential to keep the UAV engine in optimal condition and ensure safety. Always perform maintenance on a fully cooled engine. Key maintenance tasks include:

- Visual Inspection: Check for loose fittings, oil leaks, corrosion, and secure mounts (before and after each flight).
- Propeller and Mounts: Inspect for cracks, balance, and secure bolts (before each flight).
- Fuel System: Check fuel lines, connections, and clean the fuel filter (before each flight).
- Cooling Fins: Remove dust and ensure passages are unobstructed (every 20 flight hours).
- Bearings: Inspect for defects and wear (every 20 flight hours).
- Port Openings: Clean carbon deposits to ensure proper gas flow (every 40 flight hours).

- Piston and Cylinder: Inspect for corrosion and leaks at seals or gas fill valves (every 150 flight hours).

Following this schedule helps prevent engine damage, maintain performance, and reduce structural stress on the UAV.

3. Engine and Porting Calculations

Engine Sizing

This section presents the detailed calculations for engine sizing, derived from performance specifications and established engineering formulas. The engine is designed as a **short stroke** or **"oversquare" engine**, characterized by a bore-to-stroke ratio optimized for high-speed, high-load operating conditions.

Key Performance Parameters and Design Equations:

- **Bore to Stroke Ratio:** 1.2:1 (Heywood cites that typical IC engine bore/stroke ratios range from 0.8 to 1.2 for small to medium sized IC engines) (Rowton, 2014)
- **Mean Effective Pressure (P):** 43 lb/in² (Shown in the performance curves in appendix)
- **Target Engine Speed (N):** 12,000 rpm
- **Target Brake Horsepower:** 0.06 bhp

Using these parameters and formulas, the calculated engine dimensions are:

Basic Formula for Engine Performance:

$$\text{BHP} = \frac{P \times L \times A \times N}{396{,}000}$$

Rearrange Formula:

$$L \times A = \frac{396{,}000(0.06)}{43(12{,}000)} = 0.0460 \text{ in}^3 \rightarrow 0.0460 \text{ in}^3 = 754.62 \text{ mm}^3$$

Area and Bore-Stroke Relation (where D= Bore, L= Stroke):

$$A = \frac{\pi D^2}{4} \rightarrow L \times \frac{\pi D^2}{4} = 754.62 \rightarrow D = \left(\frac{754.62(4)}{\pi L}\right)^{1/2}$$

Given bore-to-stroke ratio = **1.2:1**

$$\text{D} = \left(\frac{754.62(4)}{\left(\frac{D}{1.2}\right)\pi}\right)^{\frac{1}{2}} \rightarrow \text{D} = \left(\frac{754.62(4)(1.2)}{\pi}\right)^{\frac{1}{3}} = 10.49mm$$

$$L = \frac{D}{1.2} = \frac{10.49}{1.2} = 8.74mm$$

- Stroke Length (L): 8.74 mm
- Bore Diameter (D): 10.49 mm

Engine Displacement Calculation

Engine displacement (V_d) quantifies the swept volume of the cylinder as the piston travels from Top Dead Center (TDC) to Bottom Dead Center (BDC). It is a critical factor influencing engine output and fuel consumption.

Formula for Displacement: $\boldsymbol{V_d = L \times A = L \times \left(\frac{\pi D^2}{4}\right)} \rightarrow 8.74 \times \frac{\pi(10.49)^2}{4} = 755.36 \text{ mm}^3 = approx\ 0.76 \text{ cc}$

Resulting Engine Displacement: ($V_d = 755.36\ mm^3 - approx\ 0.76\ cc$)

Clearance Volume and Compression Ratio Determination

The clearance volume (V_c), representing the volume remaining at Top Dead Center (TDC), is integral to determining the engine's compression ratio (CR).

Defined Compression Ratios:

Starting CR: 8:1 (University of Birmingham, 2025)

Running CR: 18:1 (University of Birmingham, 2025)

The height of the clearance volume (h_2) and the clearance volume (V_c) itself were computed using the following relationships:

Height of Clearance Volume (h_2): $h_2 = \frac{V_d}{(CR-1)\times A}$

$$h_{2_{starting}} = \frac{755.36}{(8-1)\times\frac{\pi(10.49)^2}{4}} = 1.249\text{ mm} \rightarrow h_{2_{starting}}: 1.249\text{ mm}$$

$$h_{2_{running}} = \frac{755.36}{(18-1)\times\frac{\pi(10.49)^2}{4}} = 0.514\text{ mm} \rightarrow h_{2_{running}}: 0.514\text{ mm}$$

Clearance Volume (V_c): $V_c = h_2\left(\frac{\pi D^2}{4}\right)$

$$V_{C_{starting}} = 1.249 \times \frac{\pi(10.49)^2}{4} = 107.945\text{ mm}^3 \rightarrow V_{c_{starting}}: 107.945\text{ mm}^3$$

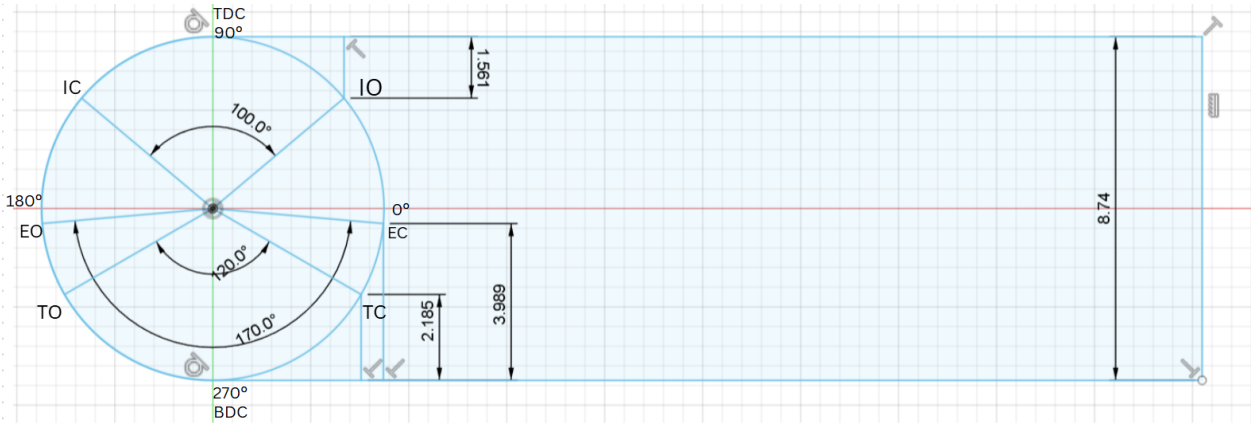
$$V_{C_{running}} = 0.514 \times \frac{\pi(10.49)^2}{4} = 44.423\text{ mm}^3 \rightarrow V_{c_{running}}: 44.423\text{ mm}^3$$

Engine Timing and Port Geometry Analysis

The precise timing of intake, exhaust, and transfer port actuation, along with their physical dimensions (heights and widths), significantly governs engine efficiency, scavenging effectiveness, and charge transfer dynamics.

Engine Timing Specifications (Crank Angles):

- Intake Port: Opens at 20°, Closes at 120° (Testbook, 2025)
- Exhaust Port: Opens at 170°, Closes at 340° (Hariharan, 2011)
- Transfer Port: Opens at 200°, Closes at 320° (Hariharan, 2011)



The diagram above was made with fusion 360. All lengths in unit millimeter (mm).

Port Heights:

- Intake Height (h_{intake}): 0.1561 cm
- Exhaust Height ($h_{exhaust}$): 0.3989 cm
- Transfer Height ($h_{transfer}$): 0.2185 cm

The percentage of the piston stroke taken up by each port is calculated using the formula:

$$\text{Percentage} = \left(\frac{\text{Port Height}}{\text{Stroke Length}}\right) \times 100$$

Using this:

Intake port percentage: $\frac{1.561}{8.74} \times 100 = 17.9\%$

Exhaust port percentage: $\frac{3.989}{8.74} \times 100 = 45.6\%$

Transfer port percentage: $\frac{2.185}{8.74} \times 100 = 25.0\%$

This shows how much of the total stroke each port occupies based on its height relative to the full piston movement.

Engine Port Widths

Port widths were systematically calculated. This involved first determining the Mean Port Height (MPH) and Mean Port Area (MPA) for each port, factoring in port duration and specific time area values. The formulae employed for calculating port width is:

We know that:

The mean port height (**MPH**) for each port is given by:

$$MPH_{\text{intake}} = 0.35 \times h_{\text{intake}} = 0.35(0.1561) = 0.0546 \text{ cm}$$
$$MPH_{\text{exhaust}} = 0.3 \times h_{\text{exhaust}} = 0.3(0.3989) = 0.120 \text{cm}$$
$$MPH_{\text{transfer}} = 0.25 \times h_{\text{transfer}} = 0.25(0.2185) = 0.0546 \text{ cm}$$

(University of Birmingham, 2025)

Port Duration Timing:

$$\text{Port Duration Time (s)} = \frac{\theta}{6N}$$
$$t_{\text{intake}} = \frac{100}{6(12{,}000)} = 0.00139 \text{ seconds}$$
$$t_{\text{exhaust}} = \frac{170}{6(12{,}000)} = 0.00236 \text{ seconds}$$
$$t_{\text{transfer}} = \frac{120}{6(12{,}000)} = 0.00167 \text{ seconds}$$

Mean Port Area (cm^2):

$$\textit{Mean Port Area} = \frac{V_d \times STA}{t}$$
$$\textit{Since } V_d = 0.76\ cm^3\textit{:}$$
$$\textit{Mean Port Area}_{intake} = 0.0820\ cm^2$$
$$\textit{Mean Port Area}_{exhaust} = 0.0467\ cm^2$$
$$\textit{Mean Port Area}_{transfer} = 0.0410\ cm^2$$

(STA and Mean Port Area calculations shown in appendix)

Port Width Calculation

To calculate the port width for each port:

$$\text{Port width} = \frac{\text{Mean Port Area(cm}^2\text{)}}{\text{Port Height(cm) - Mean Port Height(cm)}}$$ **(University of Birmingham, 2025)**

$$Port\ width_{intake} = \frac{0.0820}{0.1561 - 0.0546} = 0.808\ cm$$
$$Port\ width_{exhaust} = \frac{0.0467}{0.3989 - 0.105} = 0.159\ cm$$
$$Port\ width_{transfer} = \frac{0.0410}{0.2185 - 0.0546} = 0.250\ cm$$

Fuel Quantity Estimation for Flight Operations

Accurately estimating the required fuel quantity is paramount for optimizing the fuel-air ratio, enhancing engine performance, and ensuring fuel efficiency. This process involves calculating the total mass of fuel necessary for the specified flight duration.

To calculate the fuel volume V_{fuel}, we use:

$$V_{\text{fuel}} = \frac{M_{\text{fuel}}}{\rho_{\text{fuel}}}$$

4. Mass of fuel for a 15-minute flight:

$$M_{fuel/min} = 6.369 \times 10^{-4}\ (Appendix)$$

$$M_{\text{fuel}} = M_{fuel/min} \times 15 = 6.369 \times 10^{-4} \times 15 = 0.009553\ \text{kg}$$

5. Fuel volume (using fuel density $\rho = 0.85$ kg/L):

$$V_{\text{fuel}} = \frac{0.009553}{0.85} = 0.0112\ \text{L} = \boxed{11.2\ \text{mL}}$$

Total Fuel Capacity: 11.2 mL.

4. DESIGN CALCULATIONS

4.1 Shaft analysis

The two-stroke compression ignition engine is modelled using an idealised Otto cycle where process (1–2) represents compression and processes (3–4) represent combustion and expansion. This model is used to estimate the peak cylinder pressure so the combustion force on the piston can be found and the crankshaft and bearings can be designed to handle the maximum load.

Because process 1–2 is treated as an isentropic compression, the term pV^{γ} stays constant throughout the stroke. This allows the compressed pressure p_2 to be found using the relation:

$$p_1 V_1^{\gamma} = p_2 V_2^{\gamma}$$

Here, $\gamma = 1.4$ is the adiabatic index for air, V_1 is the trapped volume before compression, and V_2 is the clearance volume at TDC. For this engine, $V_1 = 799.78mm^3$, $V_2 = V_{c,\text{running}} = 44.42mm^3$ and the initial pressure is atmospheric, $p_1 = 0.1013$ MPa. Substituting these values gives a compressed pressure of: $p_2 = 5.80$ MPa

The isentropic temperature-pressure relation is given by: $\frac{p_1}{p_2} = (\frac{T_1}{T_2})^{\frac{\gamma}{\gamma-1}}$

Rearranging for T_2: $T_2 = T_1(\frac{p_2}{p_1})^{\frac{\gamma-1}{\gamma}}$

Where Initial temperature: $T_1 = 299.15$ K, Final compression temperature: $T_2 = 950.9$ K, Pressure ratio: $p_2/p_1 = 5.80/0.1013$. This confirms: $T_2 \approx 950.9$ K which is fully consistent with a high-compression miniature CI engine.

After compression, combustion occurs rapidly, raising both pressure and temperature inside the cylinder. Literature for miniature two-stroke compression-ignition engines reports peak combustion temperatures between 1500–2100 K (Shang et al, 2024). For this design, a peak temperature of 1600 K is selected as a realistic and safe upper-bound. To compute the corresponding peak cylinder pressure, the ideal gas proportionality is used: $\frac{p_3}{p_2} = \frac{T_3}{T_2}$

Where: $T_2 = 950.9$ K(post-compression temperature), $T_3 = 1600$ K(selected peak combustion temperature), $p_2 = 5.80$ MPa. $p_3 = p_2(\frac{T_3}{T_2})$. Thus $p_3 = 5.80(\frac{1600}{950.9}) = 9.76$ MPa

The peak force transferred to the crankshaft is: $F = p_3 A$. Where the piston area is: $A = \frac{\pi D^2}{4}$

Given: Bore $D = 10.49$ mm $= 0.01049$ m

$$A = \frac{\pi(0.01049)^2}{4} = 8.64 \times 10^{-5} \text{ m\^2}$$

$$F = 9.76 \times 10^6 \times 8.64 \times 10^{-5} = 843.5 \text{ N}$$

F= 844 N

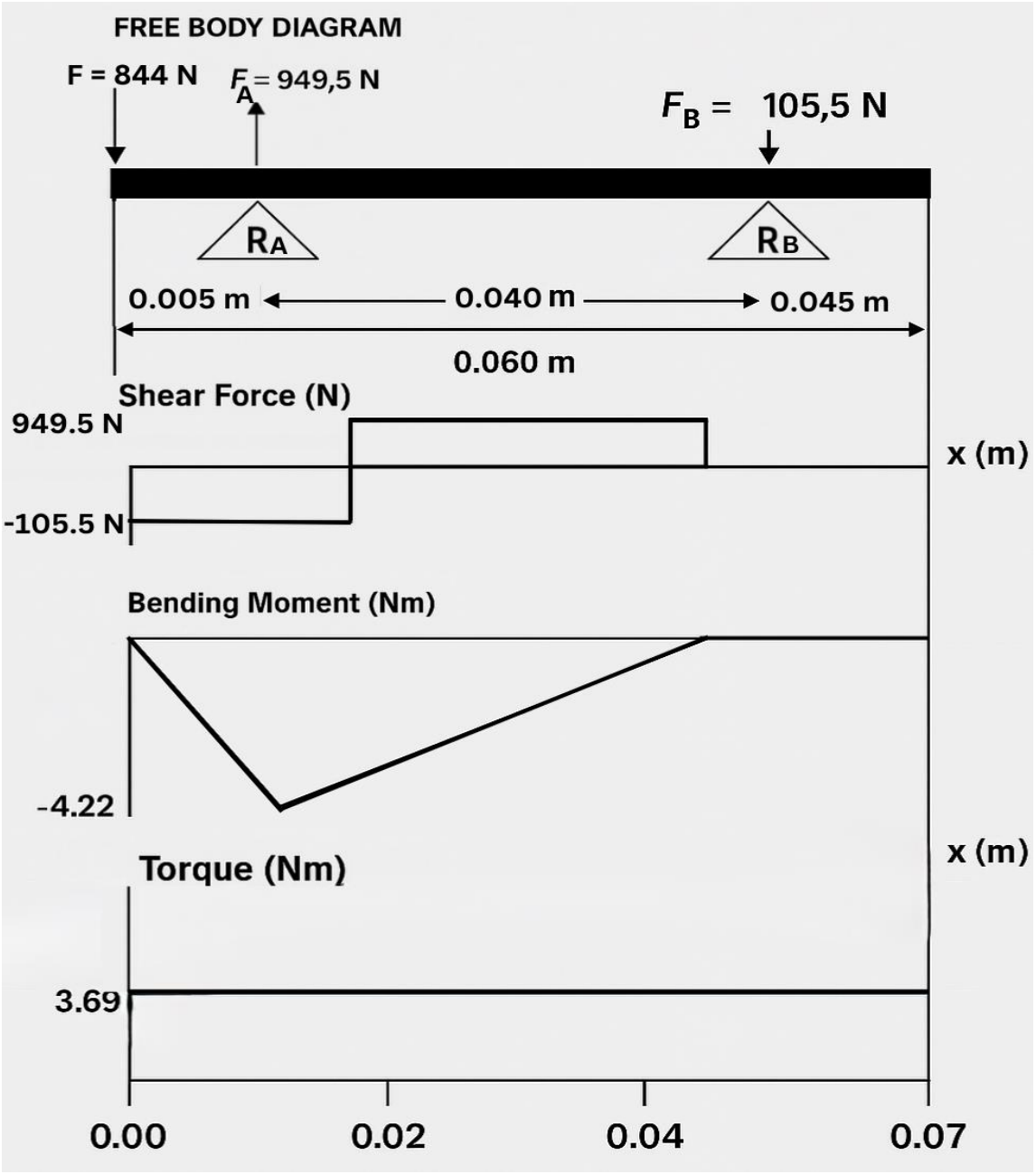
By using a free body diagram, the reactions due to the bearings on the shaft can be found. By taking moments about the applied combustion force, $\sum M_F = 0 = F_A \times 0.005 + 0.045 \times F_B$ and applying vertical equilibrium: $844 = F_A + F_B$. The bearing reactions are obtained as: $F_A = 949.5$ N, $F_B = -105.5$ N.

The maximum bending moment in the shaft occurs at the load point and is calculated as: $M_{\text{max}} = -844 \times 0.005 = -4.22$. For $0 < x < 0.005$ m: Shear force $= 844$ N and for $0.005 < x < 0.045$ m: Shear force $= 844 - 949.5 = -105.5$ N.

The torque acting on the shaft is constant and can be calculated using the equation: $T = \frac{\text{stroke}}{2} \times F$.

With stroke $L = 8.74 \times 10^{-3}$ m: $T = \frac{8.74 \times 10^{-3}}{2} * 844 = 3.69$N.

These results can be plotted on a diagram



4.2 Minimum Diameter Calculations

Material Grade	Young's modulus (GPa)	Shear modulus (GPa)	Ultimate tensile strength (MPa)	Yield stress (MPa)
AISI 4340 STEEL	210	80	1080	740

The selected material for the crankshaft is AISI 4340 steel, chosen for its excellent fatigue resistance, high toughness, and superior yield strength, making it well-suited for high-speed small-engine applications such as UAV 2-stroke engines (Shigley et al., 2015). A factor of safety (FS) = 4 was applied to account for cyclic loading and uncertainty in combustion force peaks.

Distortion Energy Theory (von Mises) Equation : $d = 3\sqrt{\frac{32}{\pi S_y} FS\sqrt{M^2 + 3/4T^2}}$

Where M = 4.22N; T = 3.69N; S_y = 740 Mpa and F_s = 2.5

$$d = 3\sqrt{\frac{32}{\pi * (740 * 10\text{^}6)} * 4 * \sqrt{(4.22)^2 + 3/4(3.69)^2}}$$

Final Minimum Diameter : $d_{\text{min}} = 6.63$ mm. Therefore, a 10mm shaft is chosen safety purposes.

4.3 Lateral Deflection

The lateral deflection of the shaft was calculated using Castigliano's theorem. The required parameters are: combustion force $P = 844$ N, distance from load to bearing A $a = 0.005$ m, distance to bearing B $b = 0.045$ m, total span $L = 0.040$ m, Young's modulus $E = 210$ GPa, and shaft diameter $d = 10$ mm. The second moment of area was first evaluated using $I = \frac{\pi d^4}{64}$, giving $I = 4.91 \times 10^{-10}$ m^4.

Using the Castigliano deflection formula $\delta = \frac{Pa^2b^2}{3EIL}$,substitution gives, $\delta = \frac{844(0.005)^2(0.045)^2}{3(210\times10^9)(4.91\times10^{-10})(0.04)}$,

resulting in the final deflection value $\delta = 3.45 \times 10^{-6}$ m, or $\delta = 3.45\ \mu$m. This deflection is within the typical acceptable range (1–10 µm) for small crankshaft applications, confirming that the selected 10 mm shaft diameter provides sufficient stiffness (Kalsi Engineering, 2019).

4.4 Steady State Loading

The steady-state normal and shear stresses in the shaft were calculated from the combined bending moment and torque. Using the relations: $\sigma_x = \frac{32M}{\pi d^3}$ and $\tau_{xy} = \frac{16T}{\pi d^3}$, with bending moment $M = -4.22$ N, torque $T = 3.69$ N, and shaft diameter $d = 10 \times 10^{-3}$ m, gives: $\sigma_x = \frac{32(-4.22)}{\pi(0.01)^3} = -4.30 \times 10^7$ Pa $= -43.0$ MPa; $\tau_{xy} = \frac{16(3.69)}{\pi(0.01)^3} = 1.88 \times 10^7$ Pa $= 18.8$ MPa

The principal stresses were then obtained from : $\sigma_{1,2} = \frac{\sigma_x}{2} \pm \sqrt{(\frac{\sigma_x}{2})^2 + \tau_{xy}^2}$,

which gives $\sigma_1 = 7.06$ MPaand $\sigma_2 = -50.0$ MPa.

Using the maximum shear stress theory, the factor of safety is : $n = \frac{S_y}{\sigma_{\max} - \sigma_{\min}} = \frac{740}{7.06-(-50.0)} \approx 13.0$

This indicates that the shaft operates well within safe limits under steady-state loading.

4.5 Torsional Deflection/ meter

The polar moment of inertia is: $J = \frac{\pi d^4}{32} = \frac{\pi(0.01)^4}{32} = 9.82 \times 10^{-10}$m^4. The torsional deflection per unit length is: $\frac{\theta}{L} = \frac{T}{JG} = \frac{3.69}{(9.82\times10^{-10})(80\times10^9)}$ = $\frac{\theta}{L} = \frac{3.69}{78.56} = 0.047$ rad/m.

This value of 0.047 rad/m is lesser than 0.05 rad/m which the upper limit for small high-speed crankshafts (Wahab, 2018). This confirms that the selected 10 mm AISI 4340 shaft provides sufficient torsional stiffness and will not experience excessive twist during engine operation.

4.6 Bearing Specification and Critical Speed Calculations



Specification	Details
Make	SKF
Model Number	608-2Z
Inner Diameter (ID)	8 mm
Outer Diameter (OD)	22 mm
Width (Thickness)	7 mm
Ball Material	Chrome Steel (AISI 52100)
Cage Material	Steel
Dynamic Load Rating	3.45 kN
Static Load Rating	1.37 kN
ABEC Rating	ABEC 5
Lubrication	Mobil DTE Oil Light (bearing oil)
Typical Price	$7.35 - $25.7 USD

SKF 608-SZ Ball Bearings (SKF,2024) 608-SZ

SKF 608-SZ deep-groove ball bearings were selected for their small 8×22×7 mm size, high fatigue strength, and reliable high-speed performance. The critical speed was evaluated using Castigliano's theorem with a shaft mass of 0.037 kg (see appendix for calculations) and bearing mass of 0.013 kg each: $w_s = 0.037g = 0.262$ N, $w_b = 0.013g = 0.127$ N. Deflection values at shaft and bearing locations were obtained using:

$\delta = (P\ a^2\ b^2)/(3\ E\ I\ L)$. With E = 210 GPa, $I = 4.91\times10^{-10}$ m⁴, L = 0.04 m, and geometric distances according to the shaft layout, the deflections were: $\delta_1 = 3.47\times10^{-9}$ m, $\delta_2 = 1.68\times10^{-9}$ m, $\delta_3 = 1.68\times10^{-9}$ m. The Rayleigh equation for the first critical speed: $n_c = (30/\pi)\ \sqrt{(\ g\ \Sigma(w\ \delta)\ /\ \Sigma(w\ \delta^2)\ )}$. Substituting values gives: $n_c = 9.55\times10^{13}$ rpm

The engine operating speed of 12,000 rpm, even with a safety factor of 1.25, is far below the calculated critical speed:12,000 × 1.25 = 15,000 rpm < 9.55×10^{13} rpm. Therefore, resonance will not occur, and the shaft–bearing system is confirmed safe for high-speed operation.

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13. University of Birmingham (2025) *Engine Displacement 2/5*. Available at: https://canvas.bham.ac.uk/courses/83227/pages/week-4-mechanical-design?module_item_id=4511079 (Accessed: 11 November 2025).

APPENDIX.

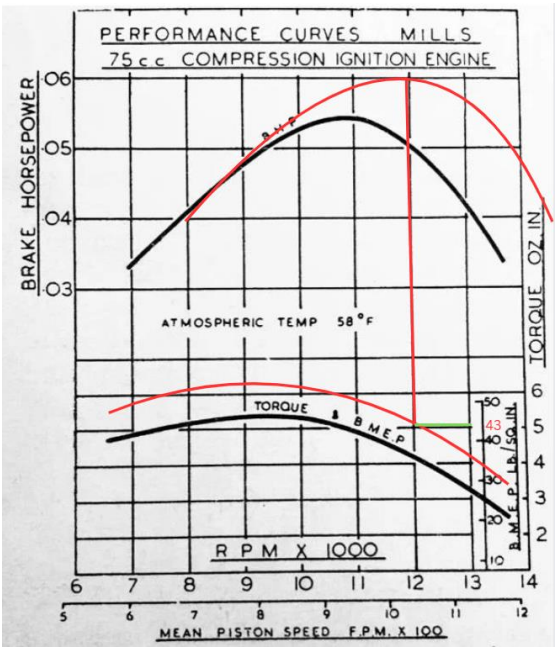
Engine Size

Target BHP: 0.06 BHP
Target Engine Speed: 12,000 rpm
Basic Formula for Engine Performance:

$$\text{BHP} = \frac{P \times L \times A \times N}{396{,}000}$$

As shown below, from the performance curve (in the provided sizing PDF), we elongated and set approximately:

$$P = 43 \text{ lb/in}^2$$



(3)

Rearrange Formula

$$L \times A = \frac{396{,}000 \times \text{BHP}}{P \times N}$$

$$L \times A = \frac{396{,}000(0.06)}{43(12{,}000)} = 0.0460 \text{ in}^3$$

Convert:

$$0.0460 \text{ in}^3 = 754.62 \text{ mm}^3$$

Area and Bore-Stroke Relation

$$A = \frac{\pi D^2}{4}$$

(where D= Bore, L= Stroke)

$$L \times \frac{\pi D^2}{4} = 754.62$$

$$D = (\frac{754.62(4)}{\pi L})^{1/2}$$

Given bore-to-stroke ratio = **1.2:1**

$$\mathrm{D} = \left(\frac{754.62(4)}{(1.2D)\pi}\right)^{\frac{1}{2}}$$

$$\mathrm{D} = \left(\frac{754.62(4)}{1.2\pi}\right)^{\frac{1}{3}}$$

$$\mathrm{D} = 9.92mm$$

$$L = \frac{D}{1.2} = 7.74\ \mathrm{mm}$$

$$L = \frac{9.29}{1.2} = 7.74\ \mathrm{mm}$$

$$L = 7.74\ \mathrm{mm}$$

Engine Displacement

$$V_d = L \times A = L \times \frac{\pi D^2}{4}$$

$$V_d = 7.74 \times \frac{\pi(9.29)^2}{4} = 524.64\ \mathrm{mm}^3 = 0.525\ \mathrm{cc}$$

Compression Ratio

$$h_2 = \frac{V_d}{(CR-1) \times \frac{\pi D^2}{4}}$$

$$h_{2_{starting}} = \frac{524.64}{(8-1) \times \frac{\pi(9.29)^2}{4}} = 1.106\ \mathrm{mm}$$

$$h_{2_{starting}} = 1.106\ \mathrm{mm}$$

$$h_2 = \frac{524.64}{(18-1) \times \frac{\pi(9.29)^2}{4}} = 0.455\ \mathrm{mm}$$

$$h_2 = 0.455\ \mathrm{mm}$$

Clearance Volume

$$V_c = h_2 \times \frac{\pi D^2}{4}$$

$$V_{C_{starting}} = 1.106 \times \frac{\pi(9.29)^2}{4} = 74.968\ \mathrm{mm}^3$$

$$V_{C_{running}} = 0.455 \times \frac{\pi(9.29)^2}{4} = 30.841\ \mathrm{mm}^3$$

Engine Port Widths

From reference (University of Birmingham, 2025), we know:
The mean port height (**MPH**) for each port is given by:

- **Intake Port:**

$$MPH_{\text{intake}} = 0.35 \times h_{\text{intake}} = 0.35(0.1561) = 0.0546\ \mathrm{cm}$$

- **Exhaust Port:**

$$MPH_{\text{exhaust}} = 0.3 \times h_{\text{exhaust}} = 0.3(0.3989) = 0.120\text{cm}$$

- **Transfer Port:**

$$MPH_{\text{transfer}} = 0.25 \times h_{\text{transfer}} = 0.25(0.2185) = 0.0546\ \text{cm}$$

Mean port area formula & Port Duration Timing

$$\text{Mean port area } (m^2) = \frac{Engine\ displacement\ (Specific\ time\ area)}{Port\ Duration}$$

$$\text{Port Duration Time (s)} = \frac{\theta}{6N}$$

where:

- θ= crank angle of each port (°)
- N= engine speed (rpm)
- **Intake Port:**

$$t_{\text{intake}} = \frac{100}{6(12{,}000)} = 0.00139 \text{ seconds}$$

- **Exhaust Port:**

$$t_{\text{exhaust}} = \frac{170}{6(12{,}000)} = 0.00236 \text{ seconds}$$

- **Transfer Port:**

$$t_{\text{transfer}} = \frac{120}{6(12{,}000)} = 0.00167 \text{ seconds}$$

Specific Time Area (STA) Reference Data

From **reference (University of Birmingham, 2025)**, the average STA values for each port type are:

Intake	0.00014 – 0.00016
Exhaust	0.00014 – 0.00015
Transfer	0.00008 – 0.00010

Values above are in $\frac{\text{s}-\text{cm}^2}{cm^3}$

Mean Specific Time Area (STA) and Mean Port Area

We take the mean of each range:

$$STA_{\text{intake}} = \frac{0.00014 + 0.00016}{2} = 0.00015\,\frac{\text{s} - \text{cm}^2}{cm^3}$$

$$STA_{\text{exhaust}} = \frac{0.00014 + 0.00015}{2} = 0.000145\,\frac{\text{s} - \text{cm}^2}{cm^3}$$

$$STA_{\text{transfer}} = \frac{0.00008 + 0.00010}{2} = 0.00009\,\frac{\text{s} - \text{cm}^2}{cm^3}$$

Mean Port Area (cm²)

$$Mean\ Port\ Area = \frac{V_d \times STA}{t}$$

Since $V_d = 0.76\ cm^3$*:*

- ***For Intake Port:***

$$Mean\ Port\ Area_{intake} = \frac{0.76(0.00015)}{0.00139} = 0.0820\ cm^2$$

- ***For Exhaust Port:***

$$Mean\ Port\ Area_{exhaust} = \frac{0.76(0.000145)}{0.00236} = 0.0467\ cm^2$$

- ***For Transfer Port:***

$$Mean\ Port\ Area_{transfer} = \frac{0.76(0.00009)}{0.00167} = 0.0410\ cm^2$$

Fuel Quantity Estimation for Flight Operations

Essential Data Inputs:

- Air Fuel Ratio (AFR): $\frac{18}{1}$ (Cantore, Mattarelli and Rinaldini, 2014)
- Air Density (p_{air}): 1.257 kg/m³
- Fuel Density (p_{fuel}) 0.85 kg/L (Filo, 2023)
- Engine Speed (N): 12,000 rpm
- Engine Displacement: 0.76 cc = $0.76\ \times 10^{-6}\ m^3$
- Required Flight Time: 15 minutes

1. Mass of air per stroke:

$$M_{\text{air}} = \text{Engine displacement} \times \rho_{\text{air}} = 0.76 \times 10^{-6} \times 1.257 = 9.553 \times 10^{-7}\ \text{kg}$$

2. Mass of fuel per stroke (assuming air-fuel ratio of 18:1):

$$M_{\text{fuel, stroke}} = \frac{9.553 \times 10^{-7}\ \text{kg}}{18} = 5.307 \times 10^{-8}\ \text{kg}$$

3. Mass of fuel per minute at 12,000 rpm:

$$M_{\text{fuel/min}} = 5.307 \times 10^{-8} \times 12000 = 6.369 \times 10^{-4}\ \text{kg}$$

Mass of Ball Bearing Calculation

Instead of starting with volume directly, compute the circular area first: $A = \frac{\pi d^2}{4}$

$$A = \frac{\pi(0.01)^2}{4} = 7.854 \times 10^{-5}\ m\text{^}2$$

Shaft volume using $V = A \times L$

$$V = 7.854 \times 10^{-5} \times 0.06$$
$$V = 6.86 \times 10^{-7}\ \ m\text{^}3$$

Mass of the shaft $m = \rho V$

$$m = 7850 \times 6.86 \times 10^{-6}$$
$$m = 0.037 \text{ kg}$$

Final Shaft Mass (AISI 4340) : M = 0.054kg